QUBEE LANG

Afaan Oromoo Beginner Programming Language

Course: Programming Language

Section - 1

Members:

1.segni nadew ugr/31205/15

2.deressa fufa ugr//15

1. LANGUAGE SPECIFICATION

1.1 Overview

QubeeLang is a beginner-friendly programming language designed using Afaan Oromoo keywords. The language translates source programs into executable Python code. The main goal of the project is to help beginners learn programming concepts using familiar local language syntax instead of English keywords.

QubeeLang is implemented as a source-to-source translator:

- Source Language: QubeeLang (`.yl`)
- Target Language: Python (`.py`)

The translator performs:

- Lexical analysis
- Parsing
- AST (Abstract Syntax Tree) generation
- Python code generation

1.2 Design Goals

The language was designed with the following goals:

- Use Afaan Oromoo keywords
- Reduce beginner confusion
- Keep syntax simple and readable
- Support basic programming concepts
- Generate readable Python code
- Help students learn programming logic easily

The language intentionally avoids advanced features such as classes and complex type systems in order to remain beginner-friendly.

1.3 Keywords and Meanings

Keyword	Meaning
`haaraa`	Declare a variable
`yoo`	If condition
`yookan`	Else condition
`yeroo`	While loop
`hojii`	Function definition
`deebi`	Return statement
`xumur`	End block
`maxxansi`	Print/output

1.4 Syntax Rules

Variable Declaration

haaraa x = 10

Assignment

x = x + 1

Arithmetic Operators

Supported operators:

- `+`
- `-`
- `*`
- `/`

Example:

haaraa z = x + y

Comparison Operators

Supported operators:

- `==`
- `<`
- `>`

Example:

```
yoo x > 5:
```

1.5 Control Flow

If / Else Statement

```
haaraa x = 10

yoo x > 5:
  maxxansi("Guddaa dha")
yookan:
  maxxansi("Xiqqaa dha")
xumur
```

While Loop

```
haaraa n = 3

yeroo n > 0:
  maxxansi(n)
  n = n - 1
xumur
```

1.6 Functions

Functions are declared using the keyword `hojii`.

Example:

```
hojii ida(a, b):
  deebi a + b
```

xumur

haaraa r = ida(2, 3)
maxxansi(r)

1.7 Beginner-Friendly Design

QubeeLang was specifically designed for beginners.

Features that help beginners include:

- Afaan Oromoo keywords
- Simple and readable syntax
- Explicit block ending using `xumur`
- Easy-to-understand structure
- Readable generated Python code
- Beginner-friendly error messages

Many students struggle with English-based programming languages. QubeeLang reduces this difficulty by allowing students to learn programming concepts using familiar words.

1.8 Translation to Python

QubeeLang programs are translated into Python code.

QubeeLang	Python
`yoo`	`if`
`yookan`	`else`
`yeroo`	`while`
`hojii`	`def`
`deebi`	`return`
`maxxansi`	`print`

1.9 Example Program

QubeeLang Source

```
hojii dachisi(x):
  deebi x * 2
xumur

haaraa y = dachisi(5)
maxxansi(y)
```

Generated Python

```
def dachisi(x):
  return (x * 2)

y = dachisi(5)
print(y)
```

2. TRANSLATOR IMPLEMENTATION

The QubeeLang translator is implemented in Python as a command-line source-to-source compiler.

Command

```
python translator.py program.yl
```

Translator Operations

The translator performs the following steps:

1. Lexical Analysis
2. Parsing
3. AST Generation
4. Python Code Generation

Input and Output

Input	Output
`.yl` file	`.py` file

Example

```
python translator.py hello.yl
```

Generated file:

hello.py

3. TEST PROGRAMS

Test Program 1 — Variables and Math

Source Code (`program1_basic_math.yl`)

```
haaraa x = 10
haaraa y = 5
haaraa z = x + y

maxxansi(z)
```

Generated Python Code

```
x = 10
y = 5
z = (x + y)

print(z)
```

Execution Result

```
15
```

Test Program 2 — Control Flow

Source Code (`program2_control_flow.yl`)

```
haaraa n = 3

yeroo n > 0:
  maxxansi(n)
  n = n - 1
xumur
```

Generated Python Code

```
n = 3

while (n > 0):
  print(n)
  n = (n - 1)
```

Execution Result

```
3
2
1
```

Test Program 3 — Feature Demonstration

Source Code (`program3_functions_feature.yl`)

```
hojii ida(a, b):
  deebi a + b
xumur

haaraa r = ida(4, 6)

maxxansi(r)
```

Generated Python Code

```
def ida(a, b):
  return (a + b)

r = ida(4, 6)

print(r)
```

Execution Result

```
10
```

4. SHORT REPORT

4.1 Design Decisions

QubeeLang was designed as a small educational programming language using Afaan Oromoo keywords. The language uses a structured lexer, parser, AST, and Python code generator instead of simple text replacement.

The syntax was intentionally kept simple and readable to reduce beginner confusion. Explicit block termination using `xumur` was chosen to make nested structures easier to understand.

The translator generates readable Python code so learners can compare QubeeLang syntax with Python syntax.

4.2 Why the Language Helps Beginners

Most programming languages use English keywords, which can be difficult for beginners.

QubeeLang helps beginners by:

- Using Afaan Oromoo keywords
- Providing simple syntax
- Supporting readable error messages
- Making programming concepts easier to understand

Students can focus on learning programming logic before struggling with English syntax.

4.3 Challenges

One major challenge was designing Unicode-aware lexical analysis for Afaan Oromoo identifiers and keywords.

Another challenge was building a parser that correctly handles nested blocks using `xumur`.

Generating readable Python output while preserving the structure of the original language also required careful AST and code-generation design.